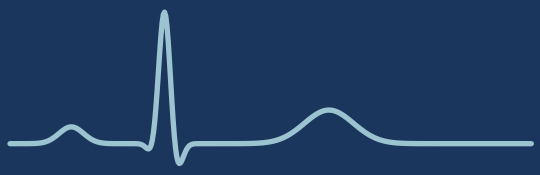
BIO 2122 | Anatomy & Physiology II | Fall 2026

Lecture 8

The Heart



Cardiac Muscle and the Conduction System

Dr. Renata Silva, Professor of Biology

Scioto Community College, North Campus

Wednesday, September 23, 2026 | Science Hall S-214

Unit 2 | Exam topic: Cardiac conduction & ECG | Unit 2 Exam: Wednesday, October 7

Notes

Tonight we move from the anatomy of the heart (Lecture 7) to its electrical activity: how cardiac muscle cells are built, how the heart starts each beat on its own, and how that activity appears on an ECG.

Everything in this deck is on the Unit 2 Exam on Wednesday, October 7. Lecture 9, the cardiac cycle and cardiac output, is on Monday, September 28.

Learning Objectives

After tonight, you should be able to:

1. Describe cardiac muscle cells and explain the role of the intercalated discs.
2. Explain the three steps of the pacemaker potential in autorhythmic cells.
3. Trace the conduction pathway in order and give the intrinsic rate of each pacemaker site.
4. Explain the contractile cell action potential, where its Ca^{2+} comes from, and why cardiac muscle cannot go into tetanus.
5. Describe how the autonomic nervous system changes heart rate.
6. Identify the waves, intervals and segments of a normal ECG, and calculate heart rate from a rhythm strip.
7. Recognize tachycardia, bradycardia, junctional rhythm, heart block and fibrillation.

Exam topic: Cardiac conduction & ECG
Objectives 1 to 7 | Unit 2 Exam, Wednesday, October 7

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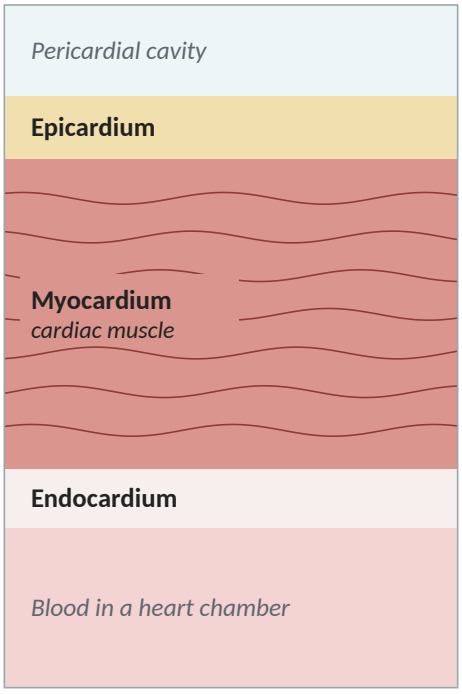
Notes

All seven objectives belong to the exam topic Cardiac conduction & ECG.

Use the terms exactly as they appear on these slides. On an exam, name a structure in full the first time, then abbreviate.

Recall from Lecture 7: The Heart Wall

- The heart wall has three layers: **epicardium**, **myocardium** and **endocardium**.
- The **myocardium** is the cardiac muscle layer. It contracts and does the pumping.
- The **fibrous skeleton** of the heart is dense connective tissue around the valves and between the atria and the ventricles. It anchors the cardiac muscle cells and the valves.
- The fibrous skeleton does not conduct impulses. It **electrically insulates** the atria from the ventricles.



The heart wall, outside to inside

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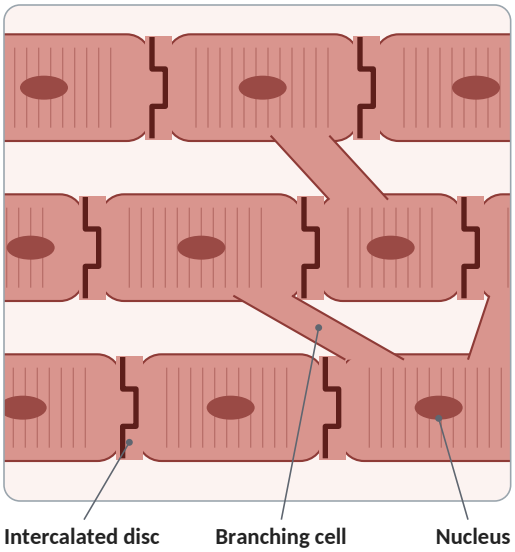
Notes

Tonight is about the myocardium: its cells, how they are connected, and how they are stimulated.

Keep the last point in mind. Because the fibrous skeleton insulates the atria from the ventricles, an impulse can reach the ventricles by only one route. We find that route on slide 9.

Cardiac Muscle Cells

- **Short, branched and striated.** Most have one nucleus; some have two.
- Joined end to end by **intercalated discs** (slide 5).
- Sarcomeres are arranged as in skeletal muscle, so the cells contract by the same sliding filament mechanism.
- **Mitochondria** fill about 25 to 35% of each cell. Cardiac muscle relies almost entirely on aerobic respiration and resists fatigue.
- T tubules are wider and fewer, and the sarcoplasmic reticulum (SR) is simpler, than in skeletal muscle.



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Notes

Compare each feature with skeletal muscle from BIO 2121; the full comparison is on slide 15.

The large number of mitochondria matters. Cardiac muscle cannot build up much of an oxygen debt, so it depends on a steady supply of oxygen from the coronary circulation.

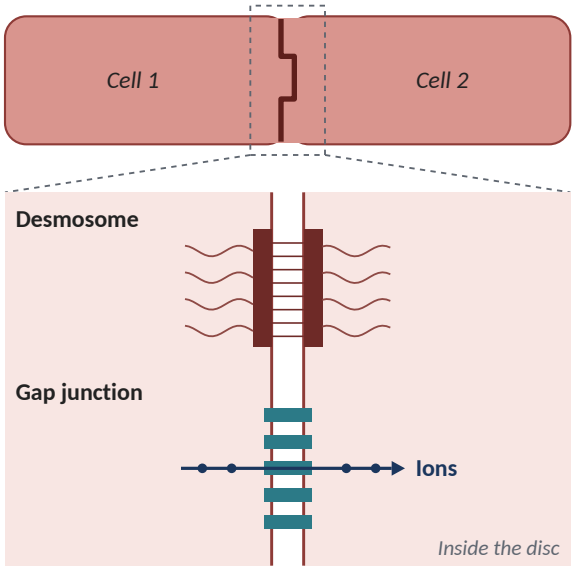
Intercalated Discs

Desmosomes

Anchoring junctions. They hold neighboring cells together, so the cells do not pull apart when they contract.

Gap junctions

Channels between neighboring cells. Ions pass directly from cell to cell, so a depolarization spreads from one cell to the next.



Result: the myocardium behaves as a **functional syncytium**. A depolarization spreads through it, and it contracts as a unit.

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Notes

A functional syncytium is a group of cells that acts as a single unit. The cells are separate, but the gap junctions connect them electrically.

Because the whole myocardium contracts as a unit, the heart does not use motor units the way skeletal muscle does (slide 15).

Two Kinds of Cardiac Muscle Cells

Contractile cells

- About **99%** of cardiac muscle cells.
- Do the mechanical work of pumping.
- Stable resting membrane potential of about -90 mV.
- Contract only when a depolarization reaches them (slide 12).

Autorhythmic cells

- About **1%** of cardiac muscle cells.
- Contract very little. They set the rhythm.
- No stable resting potential: they depolarize on their own, over and over (slide 7).
- Form the conduction system (slides 9 to 11).

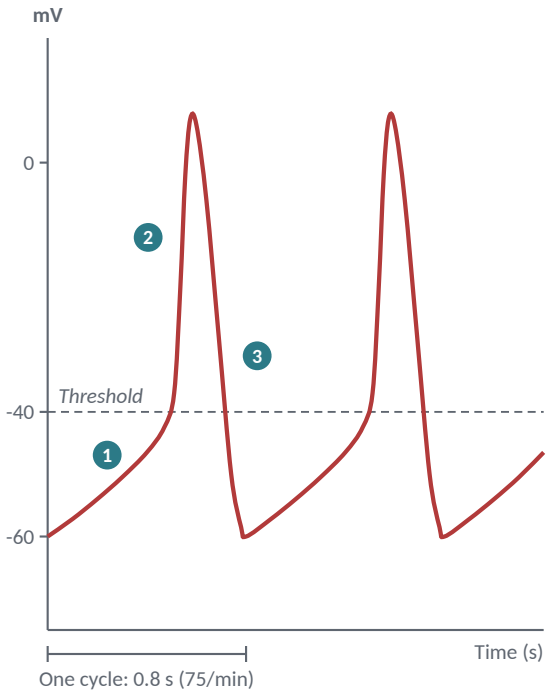
Autorhythmic cells start every heartbeat, so the heart does not need a nerve signal to contract. Nerves only speed it up or slow it down (slide 16).

Notes

Autorhythmic means self-rhythmic: these cells depolarize without any signal from a nerve. That is why a transplanted heart, which has no nerve connections, still beats.

The contractile cells do the pumping, but they wait for the depolarization that the autorhythmic cells start.

Autorhythmic Cells: The Pacemaker Potential



1. **Pacemaker potential.** Slow Na^+ inflow through open channels while fewer K^+ leave. The membrane drifts up toward threshold.
2. **Depolarization.** At threshold, about -40 mV, Ca^{2+} channels open and Ca^{2+} rushes in. This is the rising phase of the action potential.
3. **Repolarization.** Ca^{2+} channels close, K^+ channels open and K^+ flows out. The membrane returns to its most negative voltage, about -60 mV, and step 1 begins again.

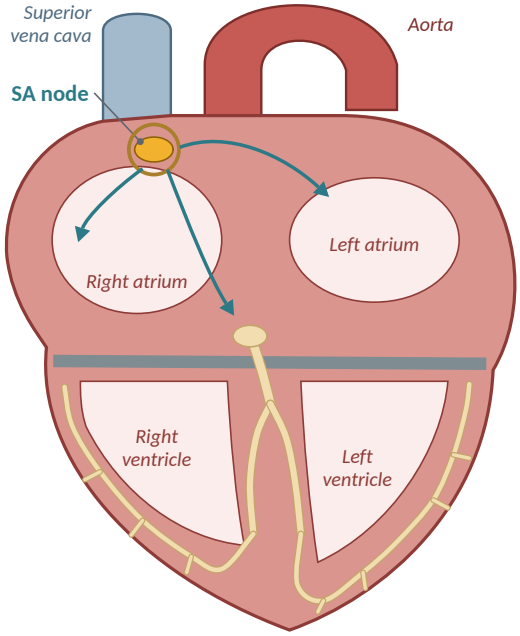
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Notes

The membrane of an autorhythmic cell never rests. As soon as it repolarizes, the next pacemaker potential begins, which is why the cycle repeats on its own.

Notice which ion carries the rising phase: Ca^{2+}. In a neuron, and in a contractile cell (slide 12), it is Na^+. The cycle on the graph takes 0.8 s, which is 75 cycles per minute, the intrinsic rate of the SA node (slide 11).

The SA Node Sets the Pace



- The **sinoatrial (SA) node** is in the right atrial wall, just below where the superior vena cava enters.
- SA node cells depolarize faster than any other autorhythmic cells, so the SA node is the **pacemaker** of the heart.
- A heartbeat started by the SA node is called **sinus rhythm**.
- The depolarization spreads through gap junctions across both atria, so the atria contract together. It also reaches the AV node (slide 9).

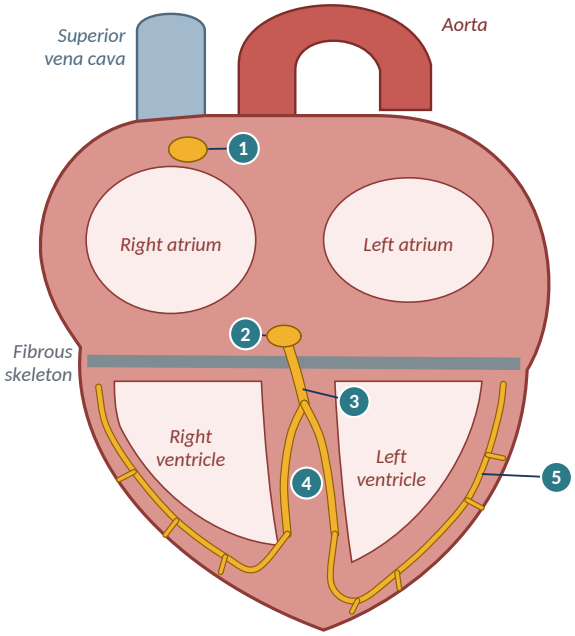
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Notes

Sinus rhythm means the SA node is the pacemaker. We use the term again when we read ECG strips (slides 22 to 25).

From the SA node, the impulse spreads cell to cell through gap junctions, so both atria depolarize and contract together. The arrows on the diagram show that spread.

The Conduction Pathway



1. **Sinoatrial (SA) node**: the pacemaker, in the right atrial wall
2. **Atrioventricular (AV) node**: in the interatrial septum, just above the tricuspid valve
3. **Atrioventricular (AV) bundle**: in the superior part of the interventricular septum
4. **Right & left bundle branches**: down the interventricular septum toward the apex
5. **Subendocardial conducting network**: from the apex up into the ventricular walls

SA node → AV node → AV bundle → right & left bundle branches → subendocardial conducting network

The **AV bundle** is the only electrical connection between the atria and the ventricles. The fibrous skeleton insulates the rest.

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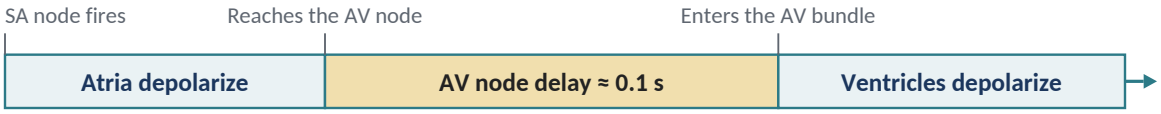
Notes

Learn the pathway in order: SA node, AV node, AV bundle, right and left bundle branches, subendocardial conducting network. These are the terms I use on exams, so use them in your answers.

The fibrous skeleton (slide 3) blocks every other route between the atria and the ventricles. That is why damage to the AV node can slow or stop conduction to the ventricles (slide 26).

Why the Impulse Pauses at the AV Node

- Conduction slows at the AV node. The **AV node delay** is about **0.1 s**.
- Why: AV node fibers are **small in diameter** and have **few gap junctions**, so the impulse crosses them slowly.
- What the delay is for: the **atria finish contracting**, and the ventricles finish filling, before the ventricles contract.
- After the AV node, conduction is fast. The AV bundle, bundle branches and subendocardial conducting network carry the impulse through the ventricles almost at once, starting at the apex.



Slow through the AV node, fast through the AV bundle, bundle branches and subendocardial conducting network

Notes

Without the AV node delay, the atria and the ventricles would contract at almost the same time, and the ventricles would contract before they had finished filling.

On the ECG, this delay is the flat line between the P wave and the QRS complex (slide 21).

Intrinsic Rates: The Fastest Pacemaker Wins

Structure	Intrinsic rate	If it paces the heart
SA node	75/min	Sinus rhythm: the normal pacemaker
AV node	50/min	Junctional rhythm (slide 25)
AV bundle and subendocardial conducting network	about 30/min	Too slow to maintain adequate circulation (slide 26)



Each autorhythmic site depolarizes on its own, at its own intrinsic rate. The SA node reaches threshold first, so it sets the heart rate. If it fails, the next fastest site takes over.

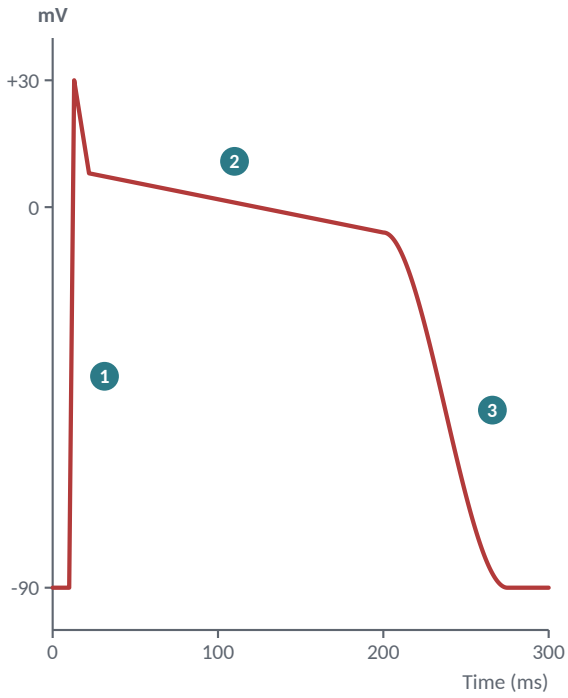
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Notes

Every autorhythmic cell can depolarize on its own, but each site has its own intrinsic rate. The SA node reaches threshold first, and its impulse resets the slower sites before they get there.

If the SA node fails, the AV node takes over at about 50/min (junctional rhythm, slide 25). The AV bundle and subendocardial conducting network beat at only about 30/min.

Contractile Cells: The Action Potential



1. **Depolarization.** Voltage-gated fast Na^+ channels open and Na^+ rushes in. The membrane goes from about -90 mV to about +30 mV in a few milliseconds.
2. **Plateau.** Slow Ca^{2+} channels open. Ca^{2+} inflow balances K^+ outflow, so the membrane stays near 0 mV for about 200 ms.
3. **Repolarization.** Ca^{2+} channels close, K^+ channels open and K^+ flows out. The membrane returns to its resting potential, about -90 mV.

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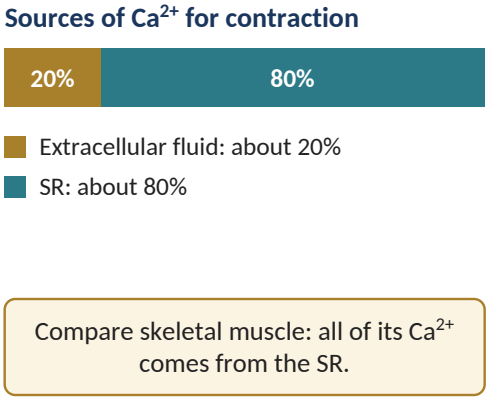
Notes

Compare this graph with the pacemaker potential on slide 7. A contractile cell has a stable resting potential, about -90 mV, and depolarizes only when the impulse reaches it.

The plateau is the key difference from skeletal muscle. Slow Ca^{2+} channels keep the membrane depolarized for about 200 ms, and that long action potential is the reason for the long refractory period on slide 14.

Where the Ca^{2+} for Contraction Comes From

1. The action potential spreads along the sarcolemma and down the T tubules.
2. Slow Ca^{2+} channels open, and Ca^{2+} enters from the extracellular fluid: **about 20%** of the Ca^{2+}.
3. That Ca^{2+} triggers the SR to release more Ca^{2+} (Ca^{2+}-induced Ca^{2+} release): **about 80%** of the Ca^{2+}.
4. Ca^{2+} binds troponin, and the filaments slide as in skeletal muscle.
5. Relaxation: Ca^{2+} is pumped back into the SR and out of the cell.



Compare skeletal muscle: all of its Ca^{2+} comes from the SR.

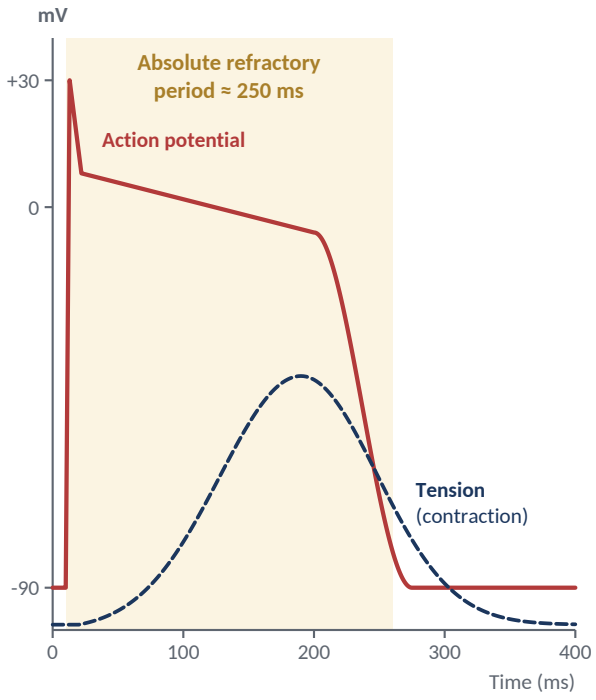
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Notes

About 20% of the Ca^{2+} enters from the extracellular fluid through the slow Ca^{2+} channels, and that Ca^{2+} triggers the SR to release the other 80%.

This is another difference from skeletal muscle, which uses Ca^{2+} from the SR only. How much Ca^{2+} enters affects how strongly the heart contracts; we return to contractility in Lecture 9.

The Absolute Refractory Period



Contractile cell: absolute refractory period ≈ 250 ms

- During the **absolute refractory period** the cell cannot respond to another stimulus.
- In a cardiac contractile cell it lasts ≈ 250 ms, **nearly as long as the contraction** itself.
- A second contraction cannot start until the first is nearly over, so contractions cannot summate.
- **Result:** it prevents tetanic contraction. The heart relaxes and refills between beats.

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Notes

The absolute refractory period is the reason cardiac muscle cannot go into tetanus. It lasts about 250 ms, nearly as long as the contraction itself, so a second contraction cannot begin until the first one is nearly over.

In skeletal muscle the absolute refractory period is only 1 to 2 ms, so stimuli can summate into tetanus. A heart in tetanus could not relax and refill, and it could not pump.

Cardiac and Skeletal Muscle Compared

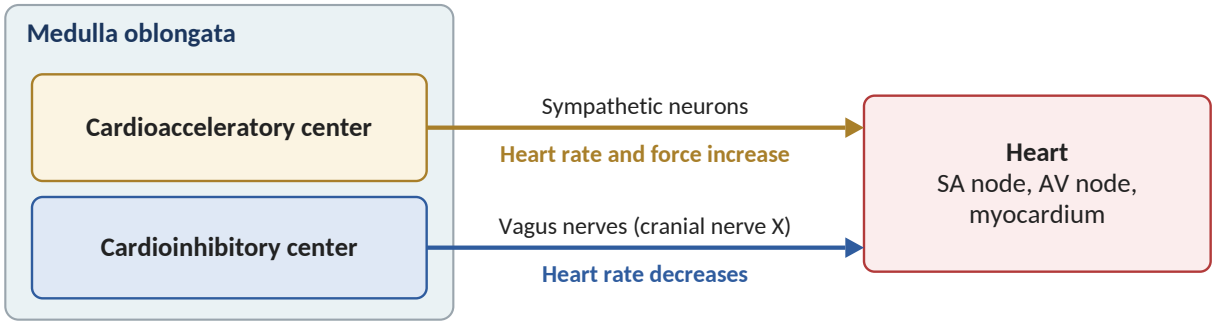
Feature	Skeletal muscle	Cardiac muscle
Cells	Long, cylindrical, many nuclei	Short, branched, usually one nucleus
Connections between cells	None. Each fiber is stimulated on its own.	Intercalated discs with desmosomes and gap junctions
What starts contraction	A motor neuron, at every fiber	Autorhythmic cells. The myocardium contracts as a unit.
Source of Ca^{2+}	SR only	About 20% extracellular fluid, about 80% SR
Absolute refractory period	1 to 2 ms	≈ 250 ms (slide 14)
Tetanus	Possible	Not possible
Energy	Aerobic and anaerobic	Almost entirely aerobic; many large mitochondria

Skeletal muscle column: review from BIO 2121.

Notes

This table pulls together slides 4 to 14. Exam questions often ask you to explain one of these differences in a sentence, for example why cardiac muscle cannot go into tetanus.

Autonomic Control of Heart Rate



- The cardiac centers are in the **medulla oblongata**.
- **Cardioacceleratory center**: sympathetic neurons to the SA node, AV node, heart muscle and coronary arteries. Heart rate and force of contraction increase.
- **Cardioinhibitory center**: parasympathetic fibers in the vagus nerves to the SA node and AV node. Heart rate decreases.
- These nerves change the rate set by the SA node. They do not start the heartbeat.

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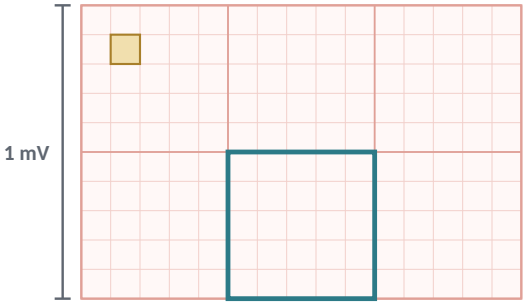
Notes

The heart beats without any nerve signal. The autonomic nervous system only speeds it up or slows it down.

The sympathetic neurons leave the spinal cord in the upper thoracic segments. The vagus nerves are cranial nerve X. Other factors that change heart rate, and the factors that change stroke volume, come in Lecture 9.

The Electrocardiogram (ECG)

- An **electrocardiogram (ECG)** is a graphic record of the electrical activity of the heart, picked up by electrodes on the skin.
- It shows the sum of all the action potentials made by nodal and contractile cells at a given time, not a single action potential.
- A standard **12-lead ECG** records 12 views of the same activity. A single-lead **rhythm strip** is used to check rate and rhythm.
- Three waves: **P wave**, **QRS complex**, **T wave** (slide 18).



1 small square = 1 mm = 0.04 s

1 large square = 5 mm = 0.20 s

Vertical: 10 mm = 1 mV

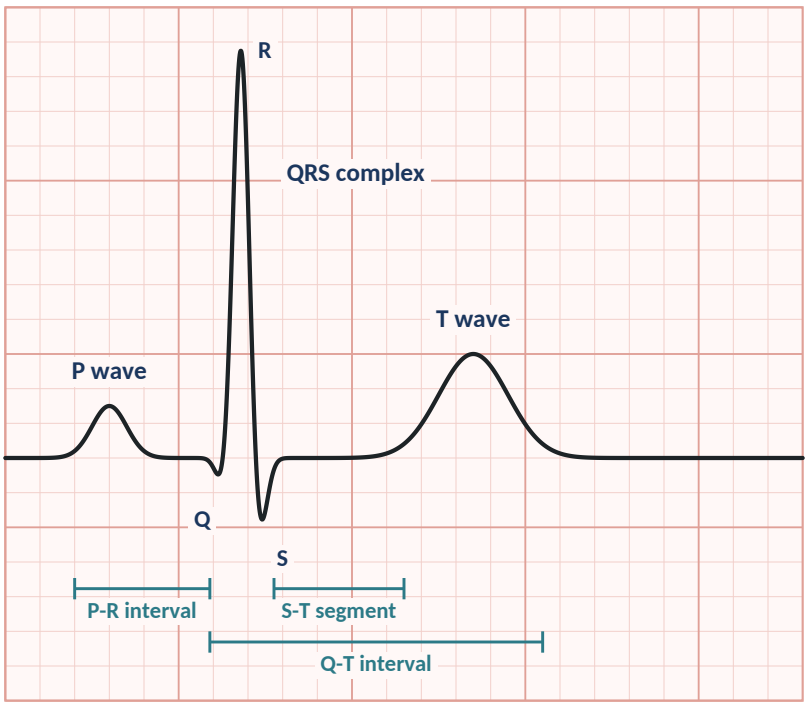
ECG paper moves at 25 mm/s.

Notes

An ECG electrode records the combined electrical activity of the whole heart, not the action potential of one cell. That is why the tracing looks so different from the graphs on slides 7 and 12.

Learn the paper scale now. You need it on slides 22 to 24 to calculate heart rate.

The Normal ECG Tracing



- **P wave**: atrial depolarization
- **QRS complex**: ventricular depolarization
- **T wave**: ventricular repolarization
- **P-R interval**: start of the P wave to the start of the QRS complex
- **S-T segment**: end of the QRS complex to the start of the T wave
- **Q-T interval**: start of the QRS complex to the end of the T wave

Drawn to scale: one small square is 0.04 s across and 0.1 mV high.

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Notes

Learn the names first: three waves, two intervals and one segment. Slides 19 and 20 explain what each one means.

The tracing is drawn to scale, so you can measure it. Count the small squares in the P-R interval: about 4, or 0.16 s.

ECG Waves: What Each One Records

Wave	What it records	What happens next	Duration
P wave	Atrial depolarization, starting at the SA node	The atria contract, about 0.1 s after the P wave begins	about 0.08 s
QRS complex	Ventricular depolarization. Atrial repolarization happens at the same time and is hidden in it.	The ventricles contract	about 0.08 s
T wave	Ventricular repolarization	The ventricles relax	about 0.16 s

The QRS complex is larger than the P wave because the ventricles have much more muscle mass than the atria.

Notes

Each wave is an electrical event, and each one comes just before a mechanical event. The P wave comes before the atria contract; the QRS complex comes before the ventricles contract.

There is no separate wave for atrial repolarization. It happens during the QRS complex, and the much larger ventricular signal hides it.

ECG Intervals and Segments

Interval or segment	Measured from	What it means	Normal
P-R interval	Start of the P wave to the start of the QRS complex	Start of atrial depolarization to start of ventricular depolarization. Includes the AV node delay.	about 0.16 s
S-T segment	End of the QRS complex to the start of the T wave	The entire ventricular myocardium is depolarized (the plateau)	flat, at the baseline
Q-T interval	Start of the QRS complex to the end of the T wave	Start of ventricular depolarization through ventricular repolarization	about 0.38 s

An **interval** includes at least one wave. A **segment** is the flat line between two waves.

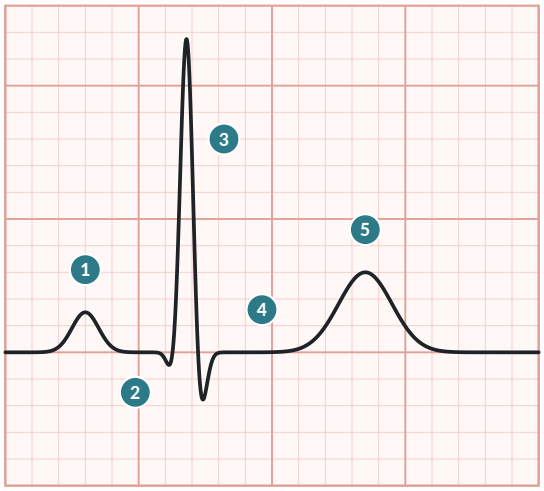
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Notes

The P-R interval includes the AV node delay (slide 10). A P-R interval longer than normal tells you that conduction through the AV node is slow.

The S-T segment matches the plateau of the contractile cell action potential (slide 12), when the whole ventricular myocardium is depolarized.

One Heartbeat: From the Pathway to the ECG



Lead II, 25 mm/s

1. SA node fires and the atria depolarize: **P wave**
2. AV node delay, about 0.1 s: the flat line after the P wave, part of the **P-R interval**
3. AV bundle → right & left bundle branches → subendocardial conducting network. The ventricles depolarize from the apex up: **QRS complex**
4. The ventricles contract during the plateau: **S-T segment**
5. The ventricles repolarize: **T wave**

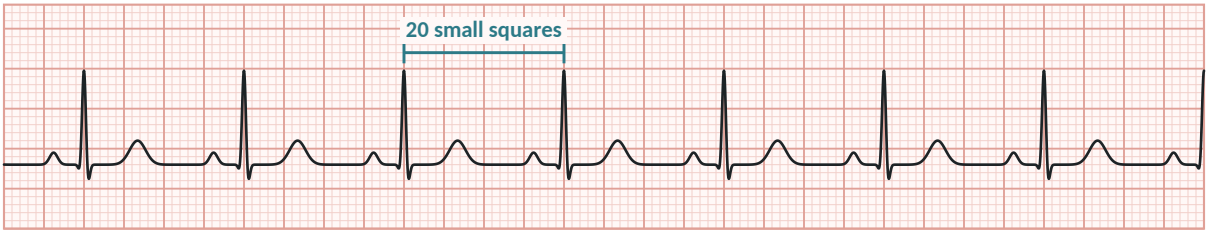
The atria repolarize during the QRS complex, so there is no separate wave for it.

Notes

Read this slide with slide 9 beside it. Each structure in the conduction pathway produces part of the tracing.

The ventricles contract during the S-T segment, after the QRS complex, because an electrical event always comes just before the mechanical event it triggers.

Reading Heart Rate From a Strip, Step by Step



Lead II rhythm strip, 25 mm/s

Step 1. Count the small squares from one R wave to the next: **20**.

Step 2. Convert to time: 20 × 0.04 s = **0.80 s** per beat.

Step 3. Convert to rate: 60 s/min ÷ 0.80 s per beat = **75 beats/min**.

Step 4. Check: 75/min is between 60 and 100/min, so the rate is normal.

Shortcuts

Heart rate = 1500 ÷ small squares between R waves

1500 ÷ 20 = **75/min**

Heart rate = 300 ÷ large squares between R waves

300 ÷ 4 = **75/min**

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Notes

Work every strip the same way: count the small squares from one R wave to the next, convert to seconds, then convert to beats per minute.

The shortcuts work only at the standard paper speed of 25 mm/s. If the distance between R waves changes from beat to beat, count the QRS complexes in 6 seconds (30 large squares) and multiply by 10.

Try It: Two Practice Strips

Strip A

25 mm/s



Strip B

25 mm/s



For each strip:

1. Calculate the heart rate, step by step as on slide 22.
2. Is the rate normal, tachycardia or bradycardia?
3. Look for a P wave before each QRS complex. Which structure is pacing the heart?

Work both strips before you look at slide 24.

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Notes

Work both strips before you look at slide 24. Use the steps on slide 22.

For each strip, decide two things: the rate, and whether a P wave comes before every QRS complex.

Check Your Work: Strips A and B

Strip A

Step 1. 12 small squares between R waves.

Step 2. 12 × 0.04 s = 0.48 s per beat.

Step 3. 60 ÷ 0.48 = **125 beats/min**.

Step 4. Over 100/min: **tachycardia**. A P wave comes before every QRS complex, so the SA node is pacing the heart (sinus tachycardia).

Strip B

Step 1. 30 small squares between R waves.

Step 2. 30 × 0.04 s = 1.20 s per beat.

Step 3. 60 ÷ 1.20 = **50 beats/min**.

Step 4. Under 60/min: **bradycardia**. There are no P waves and the rate is 50/min, so the AV node is pacing the heart (junctional rhythm, slide 25).

Bradycardia	Normal resting rate	Tachycardia
under 60/min	60 to 100/min	over 100/min

Count from the peak of one R wave to the peak of the next.

Notes

Strip A is fast but otherwise normal. Every QRS complex has a P wave in front of it, so the SA node is still the pacemaker.

Strip B is slow and has no P waves. A rate of about 50/min with no P waves points to the AV node (slide 11).

Abnormal Rates and Rhythms

- **Tachycardia**: resting heart rate over 100/min.
- **Bradycardia**: resting heart rate under 60/min.
- **Junctional rhythm**: the SA node is not working, so the AV node paces the heart at about 50/min. There are no P waves.
- **Ectopic focus**: an abnormal pacemaker outside the SA node. If it fires before the SA node, the result is a premature contraction (extrasystole).

Normal sinus rhythm, 75/min



Junctional rhythm, 50/min: no P waves



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Notes

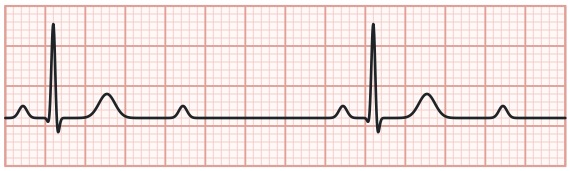
Tachycardia and bradycardia describe the rate only. Junctional rhythm describes which structure is pacing the heart.

In junctional rhythm the AV node acts as the pacemaker, so the rate falls to the intrinsic rate of the AV node, about 50/min.

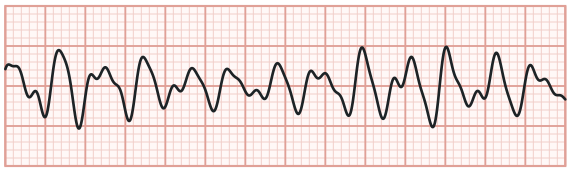
Heart Block and Fibrillation

- **Heart block**: damage to the AV node slows or stops conduction to the ventricles.
- **Partial heart block**: some atrial impulses do not reach the ventricles, so there are more P waves than QRS complexes.
- **Total heart block**: no impulses get through. The ventricles beat at their intrinsic rate, about 30/min (slide 11), too slow to maintain adequate circulation.
- **Fibrillation**: rapid, irregular, out of phase contractions. The heart cannot pump.
- **Defibrillation**: an electrical shock depolarizes the entire myocardium at once, so the SA node can take over again.

Partial heart block: 2 P waves per QRS complex



Ventricular fibrillation: no recognizable waves



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Notes

Heart block is a conduction problem; fibrillation is a rhythm problem. In ventricular fibrillation the ventricles quiver instead of contracting, and blood flow stops.

In total heart block the ventricles fall back to their own intrinsic rate from slide 11, about 30/min. An artificial pacemaker can then set the heart rate.

Summary

1. Intercalated discs join cardiac muscle cells; gap junctions make the myocardium a functional syncytium (slides 4 and 5).
2. Autorhythmic cells depolarize on their own: slow Na^+ inflow, then Ca^{2+} in, then K^+ out (slide 7).
3. Pathway: SA node → AV node → AV bundle → right & left bundle branches → subendocardial conducting network (slide 9).
4. The AV node delays the impulse about 0.1 s, so the atria finish contracting first (slide 10).
5. Intrinsic rates: SA node 75/min, AV node 50/min, AV bundle and subendocardial conducting network about 30/min (slide 11).
6. Contractile cells: fast Na^+ in, a plateau from slow Ca^{2+} in, then K^+ out. About 20% of the Ca^{2+} comes from the extracellular fluid and 80% from the SR (slides 12 and 13).
7. Absolute refractory period ≈ 250 ms, nearly as long as the contraction: no tetanus (slide 14).
8. ECG: P wave, QRS complex, T wave. P-R interval about 0.16 s; Q-T interval about 0.38 s (slides 18 to 20).

Notes

If you can explain every line on this slide in your own words, you are ready for the conduction and ECG questions on the Unit 2 Exam.

Before Monday

Practice on your own:

1. Redraw the conduction pathway from memory. Label each structure with its intrinsic rate (slides 9 and 11).
2. Label a blank ECG tracing: P wave, QRS complex, T wave, P-R interval, S-T segment and Q-T interval (slide 18).
3. Work the strips on slide 23, then check slide 24.
4. Read the textbook sections listed for this lecture on Scioto Learn.

Next class: Lecture 9
The cardiac cycle and cardiac output
Monday, September 28, 6:00 pm, Science Hall S-214

Unit 2 Exam
Wednesday, October 7, 6:00 pm
Lectures 7 to 11, plus the carry-forward section on blood and endocrine

Questions
Office hours: Monday and Wednesday, 4:30 to 5:45 pm, Science Hall S-305
rsilva@sciotocc.edu

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Notes

These tasks are for self-study. They are not collected or graded.

Bring your questions on tonight's material to office hours, Monday or Wednesday, 4:30 to 5:45 pm, in Science Hall S-305, or email me at rsilva@sciotocc.edu.